

Volume III · Chapter 12 — Retrieval-Augmented Generation (RAG) · Theory

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-III-Chapter-12-context.md for the AI interaction log.

Prerequisites: Ch 7, Ch 8, Ch 11.

Learning outcomes — the student can: explain RAG architecture; build an ingest→embed→retrieve→generate pipeline; ground answers with citations; evaluate retrieval and answer faithfulness; reduce hallucination.

Topics (theory) — 12 h:

#	Topic	h
12.1	Why RAG; grounding vs. parametric knowledge	1
12.2	RAG architecture: retriever + generator; the pipeline	2
12.3	Chunking strategies for clinical documents	2
12.4	Embeddings & similarity search fundamentals	2
12.5	Prompt assembly, context windows, citation/provenance	2
12.6	RAG evaluation:	2

#	Topic	h
	retrieval metrics + answer faithfulness	
12.7	Advanced: reranking, hybrid search, query rewriting	1

Datasets/tools: LangChain/LlamaIndex or custom; embeddings model; Chroma; local LLM. **Assessment:** working RAG system (**60%**); eval report (**20%**); quiz (**20%**). **Key decisions:** chunk size/overlap; embedding model; RAG vs. fine-tune; citation strategy. **References:** plan §13 → *RAG, vector databases & local inference*. **Hours:** Theory **12** + Lab **18** = **30**.